# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

### SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A girl of 13 has a growth spurt before boys of the same age. This is: 1

**(a)** a disease **(b)** abnormal **(c)** an illness **(d)** normal

**2.** A rod is heated at one end. The time taken for wax pins to fall off is least for the rod made of: 1

**(a)** copper **(b)** glass **(c)** iron **(d)** wood

**3.** A car covers 90 km in 1.5 h. Its speed is: 1

**(a)** 135 km/h **(b)** 45 km/h **(c)** 60 km/h **(d)** 90 km/h

**4.** Which of the following correctly shows the order of the alimentary canal? 1

**(a)** stomach, mouth, oesophagus, intestine

**(b)** mouth, oesophagus, stomach, small intestine, large intestine

**(c)** oesophagus, mouth, stomach, small intestine

**(d)** mouth, stomach, oesophagus, large intestine, small intestine

**5.** In which of the following are boys and girls same at puberty? 1

**(a)** beard growth **(b)** menstruation **(c)** voice deepening **(d)** growth spurt

**6.** During sea breeze, the pressure of air above land is: 1

**(a)** higher than over sea **(b)** zero

**(c)** lower than over sea **(d)** the same

**7.** A car travels 40 km in 1 h and then 60 km in the next hour. Its average speed is: 1

**(a)** 100 km/h **(b)** 40 km/h **(c)** 50 km/h **(d)** 60 km/h

**8.** The average number of breaths per minute for a resting adult is 16. The number in an hour is: 1

**(a)** 96 **(b)** 1600 **(c)** 960 **(d)** 160

**9.** The thyroid gland is in the: 1

**(a)** brain **(b)** chest **(c)** abdomen **(d)** neck

**10.** Which of these clothes would be best on a very hot, sunny day? 1

**(a)** woollen **(b)** black polyester

**(c)** leather **(d)** white loose cotton

**11.** In a distance-time graph, the slope gives: 1

**(a)** distance only **(b)** speed **(c)** time **(d)** acceleration

**12.** The heart of a human has: 1

**(a)** two chambers **(b)** four chambers **(c)** three chambers **(d)** five chambers

**13.** A boy's voice breaks at 14. It shows that his: 1

**(a)** lungs are growing **(b)** larynx is growing

**(c)** heart is growing **(d)** teeth are growing

**14.** Heat reaches us from a bonfire at a distance mainly by: 1

**(a)** evaporation **(b)** radiation **(c)** convection **(d)** conduction

**15.** A runner covers 1.5 km in 10 min. His speed in km/h is: 1

**(a)** 1.5 km/h **(b)** 0.15 km/h **(c)** 15 km/h **(d)** 9 km/h

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** Why do boys' voices change more than girls' at puberty? 2

**17.** Why does a metal rod conduct heat faster than a wooden stick? 2

**18.** A car travels 150 km in 2.5 h. Find its speed in km/h and m/s. 2

**19.** Why does the number of breaths per minute rise during exercise? 2

**20.** What are the benefits of regular exercise for teenagers? 2

## SECTION — C (Short Answer Type) 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** A teenager is fond of cola and chips and avoids vegetables. Explain the health risks and suggest changes. 3

**22.** Explain how an electric heater warms a room using all modes of heat transfer. 3

**23.** A cyclist covers 12 km in 40 minutes. Find the speed in km/h and m/s. 3

**24.** Trace the path of a piece of bread from mouth to anus, with changes at each stage. 3

OR

What are the factors that lead to tobacco use among teens and how to prevent it? 3

**25.** Give reasons: 3

(a) pans are made of metal and have plastic handles

(b) the tank for hot water is covered with a wool jacket.

OR

A man walks to a market 3 km away at 5 km/h and returns at 3 km/h. Find the total time. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** Describe the changes at puberty in a table-like form for boys and girls, mentioning the hormones involved. 5

OR

Describe the three experiments to demonstrate conduction, convection and radiation. 5

**27.** A car starts from city A at 8:00 a.m. and reaches city B 240 km away at 12:00 noon, with a half-hour stop. Find the average speed including the stop and the running speed. 5

OR

Explain the whole process of digestion of a meal consisting of chapati (starch), dal (protein) and ghee (fat), naming the organs, juices and the end products. 5

**END OF QUESTION PAPER** **Total: 50 Marks**

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